Fréchet Dispersion and the Exact Collar-Failure Charge

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Abstract

We identify the incident link energy with the Fréchet energy of its staple family. This corrects the deterministic collar-failure charge from $\eta^2/2$ to η^2 and introduces dispersion as the natural retained variable.

Let $G = SU(2)$ carry its bi-invariant distance and let $S_1, \dots, S_m$ be the staples incident on a link U. Define

$$\Phi(C) = \sum_i d(C, S_i)^2, \quad \mathcal{D} = \inf_C \Phi(C), \quad R = \inf_C \max_i d(C, S_i).$$

Theorem 1 (Exact identity and charges)**.** *For every U,*

$$E(U) := \sum_i d(I, US_i)^2 = \Phi(U^{-1}) \geq \mathcal{D}.$$

If $R > \eta$, then $E(U) > \eta^2$ for every U.

Proof. Bi-invariance gives $d(I, US_i) = d(U^{-1}, S_i)$. The first assertion follows by summation and minimization. If $R > \eta$, no center contains all staples in an η-ball. Taking the particular center U^{-1} gives an index with $d(U^{-1}, S_i) > \eta$; its squared contribution already exceeds η^2. □

With heat-kernel weight $\exp[-E/(2t)]$, the corrected local exponent is $\eta^2/(2t)$, twice the earlier diameter-only value $\eta^2/(4t)$. The result is gauge invariant because a common endpoint gauge action transports centers and preserves all distances.

Remark 1. *This is a deterministic action statement. Radial Haar entropy, witness multiplicity, conditional normalization, retained states, and volume uniformity remain separate. No physical basin entry follows.*